Phase 2 Report: Data Collection, Understanding and Preparation

Can Early Training Dynamics Predict Out-of-Distribution Generalization?

Project ID: F26-12

Track A

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Companion code: F26-12.ipynb (all figures, tables and checks referred to below are produced there).

● Dataset Source and Description

The project needs a controlled setting where shortcut strength can be tuned, and a realistic benchmark to validate on. We therefore use two datasets.

Dataset	Source	Description
Colored MNIST (generated)	Built by us from MNIST (LeCun et al.), following the construction of Arjovsky et al., 2019 (Invariant Risk Minimization).	Handwritten digits, 10 classes, 28x28. Each class is tied to one color, and the colour-label agreement p is a parameter. MNIST itself has 60,000 training and 10,000 test images.
Waterbirds	Sagawa et al., 2020 (Distributionally Robust Neural Networks), built from CUB-200-2011 bird photos and Places backgrounds. Obtained from the original Stanford release (also available via WILDS).	Binary task: landbird vs. waterbird, with background (land or water) as the shortcut. 11,788 images: 4,795 training, 1,199 validation, and 5,794 test.

The Kaggle Colored MNIST dataset listed in our approved proposal is a pre-generated, static copy, so it cannot supply controlled variation in p; we generate our own version and use the Kaggle copy only as a reference.

● Data Collection Method

- **MNIST:** downloaded programmatically from the public mirrors that torchvision uses (idx files), parsed with NumPy, and cached locally. No scraping or API keys are needed.
- **Colored MNIST:** synthesized. Each class gets one of 10 evenly spaced hues. With probability p an image receives the color of its true class; otherwise, one of the other nine colors uniformly. Only the digit pixels are tinted. p = 0.10 therefore means color is independent of the label (control, no shortcut), and p = 0 means color never matches (reversed). Shortcut strengths generated: p in {0.10, 0.50, 0.70, 0.90, 0.95, 0.99}.
- **Waterbirds:** downloaded as the original tarball (images plus metadata.csv with label, background, and split). Group = 2 x label + background, giving four groups: landbird/land, landbird/water, waterbird/land, waterbird/water.

All randomness is seeded (seed 0 plus a per-p offset), so the datasets are exactly reproducible.

Preprocessing Steps

Colored MNIST

- Validation checks: uint8 dtype, pixel range 0-255, all 10 labels present, no missing values. MNIST ships clean, so no samples were removed.
- Splits: the 60,000 official training images are randomly split into 50,000 train and 10,000 in-distribution validation; the 10,000 official test images are the test set.
- Color assignment is done independently per split and per p. The test set additionally has two out-of-distribution versions: **reversed** (color never matches the label) and **uncorrelated** (color independent of the label).
- Only the color index per image is stored (about a few hundred KB in total); images are tinted on the fly, so every shortcut strength shares one copy of MNIST. Pixels are scaled to [0,1] and normalized per channel using statistics from 10,000 training images.

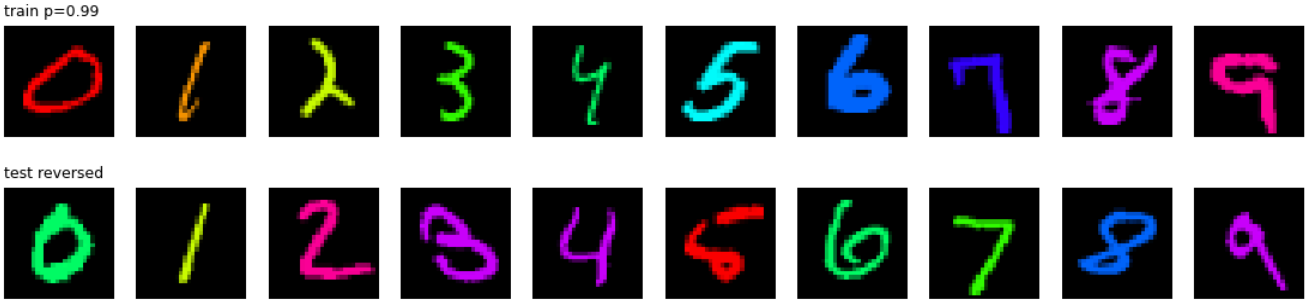
Waterbirds

- Validation checks: no missing values, unique filenames, valid label/background/split values, every listed image file exists, and total and training-group counts match the published dataset.
- Splits: the official train/validation/test split is kept. Because the official validation and test sets are balanced across backgrounds, they are already shifted relative to training. We therefore hold out 15% of each training group (about 720 images, stratified) as an **in-distribution holdout**, which the baselines (ATC, Agreement-on-the-Line) need as an in-distribution reference. About 4,075 images remain for fitting.
- Images are converted to RGB, resized to 256x256, center-cropped to 224x224, scaled to [0,1], and normalized with ImageNet mean and standard deviation.
- PyTorch Dataset classes for both datasets return (image, label, colour or group), so the same group/color information is available later for the feature-ablation reliance score and group-wise evaluation.

Brief EDA: Key Insights

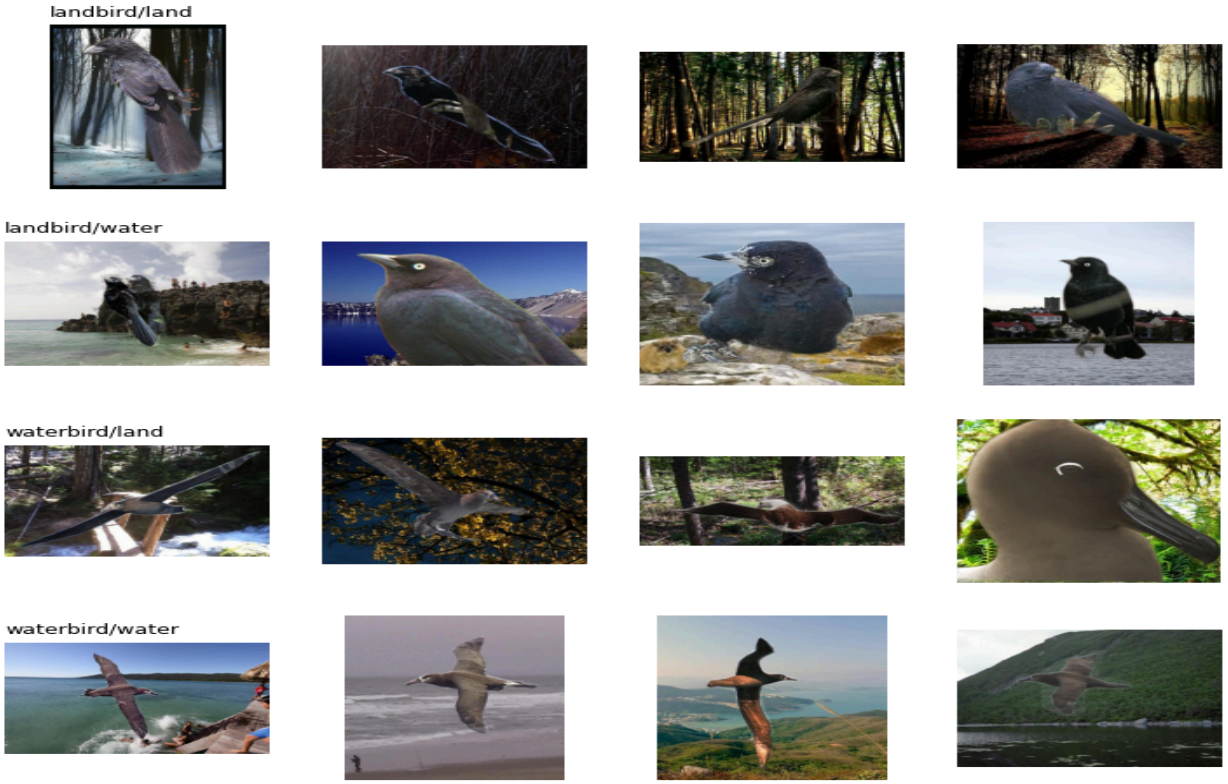
Colored MNIST

- **Classes are nearly balanced.** In the original 60,000 training images, class 1 is the most frequent (6,742) and class 5 the least (5,421), a ratio of about 1.24, so plain accuracy is a fair metric.
- **Shortcut strength matches the target.** The measured color-label agreement on every split equals the target p up to sampling error (checked with an automated tolerance in the notebook); the reversed set has agreement 0, and the uncorrelated set is about 0.10.
- **A pure shortcut model is easy to describe.** A classifier that outputs the class whose color it sees scores about p on in-distribution data, about 0.10 on the uncorrelated set and 0 on the reversed set. At p = 0.99 it looks nearly perfect in-distribution yet fails completely out-of-distribution, which is exactly the failure the project wants to forecast early.
- **The real signal is learnable without color.** A linear classifier on grayscale pixels is run as a shape-only reference (result reported in the notebook), confirming the digit shape alone carries the label, so a model that ignores color can generalize.



Waterbirds

- **Strong, deliberate shortcut.** In the training set 95.0% of waterbirds are on water backgrounds (1,057 of 1,113) and 95.0% of landbirds on land (3,498 of 3,682).
- **Class and group imbalance.** Landbirds are 76.8% of training images. Training group sizes are 3,498, 184, 56 and 1,057; the minority groups are 240 images, about 5% of training, and the smallest group (waterbird on land) has only 56 images. Worst-group accuracy will therefore be noisy, and we report confidence intervals.
- Group 0: Landbird on Land (majority group)
- Group 1: Landbird on Water (minority group)
- Group 2: Waterbird on Land (minority group)
- Group 3: Waterbird on Water (majority group)
- **Shift is built into the splits.** Unlike training, the validation and test sets are balanced across backgrounds, which is why the in-distribution holdout described above is needed. Group-level counts for all subsets are printed in the notebook.
- **Variable image sizes.** Source photos differ in width and height (summary in the notebook), which is why a fixed resize and crop is applied.
- **Limited variation.** Waterbirds has a single shortcut strength, so variation across models there must come from seeds, architectures, and training settings.



- **Final Dataset Summary**

Dataset	Subset	Images	Input shape	Notes
Colored MNIST	train	50,000	3x28x28	one version per p in {0.10, 0.50, 0.70, 0.90, 0.95, 0.99}
	validation	10,000	3x28x28	same p as training (in-distribution)
	test	10,000	3x28x28	in-distribution per p, plus reversed and uncorrelated out-of-distribution sets
Waterbirds	train (fit)	about 4,075	3x224x224	95% background-label agreement, 4 groups
	in-distribution holdout	about 720	3x224x224	15% of each training group, stratified
	validation	1,199	3x224x224	official; balanced across backgrounds
	test	5,794	3x224x224	official; out-of-distribution evaluation

Both datasets are cleaned, split, documented and wrapped in Dataset classes, ready for model training in Phase 3. All numbers in this report that depend on a run (agreement tables, group counts by subset, image-size summary, shape-only accuracy) are produced by the notebook.